Lecture 1: Introduction to Statistics and Probability

Lecturer

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CLASS CODE

TOAE203

COURSE

Statistics and Probability in Economics and Business

AGENDA

01

READING A NUMBER

Hai con số chính thống trái nhau, ba câu đố, và vì sao trực giác hỏng có hệ thống.

02

THE MEANING OF PROBABILITY

Cùng một dấu phần trăm, bốn loại nghĩa khác nhau. Bảy con số bạn đã gặp trong 24 giờ.

03

FROM DATA TO DECISION

Một dòng dữ liệu được nâng dần lên thành một quyết định.

04

WHERE DATA COMES FROM

Sơ cấp và thứ cấp, các nguồn dữ liệu Việt Nam, và chất lượng dữ liệu.

05

OPENING THE FILE

Excel và Python, bộ dữ liệu xuyên suốt môn học, và sản phẩm nộp sau buổi.

READING A NUMBER

Vietnam, 2025 — which one is wrong?

Two official numbers. Same agency, same country, same year

Home > VCCI Ecoinomics news > Vietnam's economic growth hits 8.02% in 2025, GDP per capita $5,026

VIETNAM'S ECONOMIC GROWTH HITS 8.02% IN 2025, GDP PER CAPITA $5,026

Wed, Jan 07, 2026, 15:34:00

Vietnam's economy grew 8.02% in 2025, its second-fastest pace in the past 15 years, driven mainly by services and industrial production, official data shows.

The economic expansion was second only to 2022's 8.12% over the 2011-2025 period, the General Statistics Office (GSO), under the Ministry of Finance, reported on Monday.

Source: VIETNAM CHAMBER OF COMMERCE AND INDUSTRY

ECONOMY

Per capita income climbs 9.3% in 2025

VGP - Average per capita income in 2025 was estimated at VND5.9 million (approximately US$225) per month, marking a 9.3 percent increase from 2024, according to preliminary findings of the Household Living Standards Survey 2025 conducted by the National Statistics Office (NSO).

Source: Government News

READING A NUMBER

Vietnam, 2025 — which one is wrong?

Two official numbers. Same agency, same country, same year.

	Indicator	Value	Per month
A	GDP per capita	125.5 million VND / year (≈ USD 5,026)	≈ 10.5 million VND
B	Average income per capita	from the Household Living Standards Survey	5.9 million VND
	The gap		nearly 2×

Vote now: A is wrong · B is wrong · both are wrong · I don't know

READING A NUMBER

Puzzle 1 — Bill Gates enters a bar

There are 9 men sitting in a bar, each earns $1000 a month. Suddenly, Bill Gates moves in, earning $100,000,000 a month. What is the average income in that room now — and are the 9 men any richer?

HOW IT WAS MADE

Sum the 10 incomes, divide by 10

Roughly 10 millions USD per person per month. The arithmetic is completely correct

WHAT IT CANNOT SAY

Nothing about how the 9 men live

One extreme value dragged the average away from every single person in the room.

A number can be arithmetically right and descriptively false.

READING A NUMBER

Puzzle 2 — Which restaurant do you want to go?

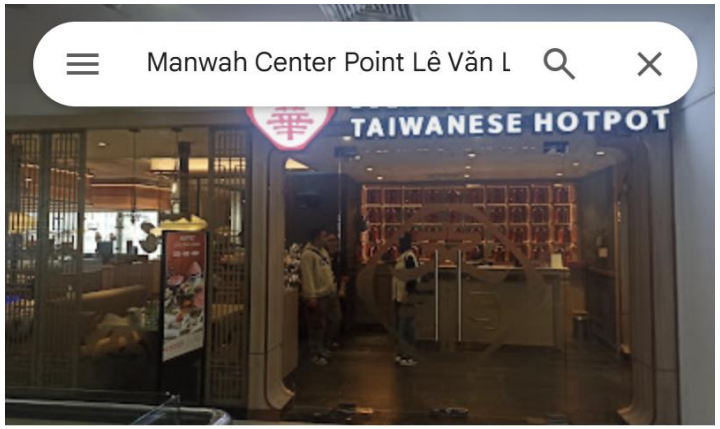
Restaurant A: 4.9 ★ from 54 ratings. Restaurant B: 4.4 ★ from 1,163 ratings.



Shan Yu Qing

4.9 ★★★★★ (54) · ₫200–300K

Chinese restaurant



Manwah Center Point Lê Văn Lương

4.4 ★★★★½ (1,163) · ₫400–500K

Hot pot restaurant

Overview Menu Reviews About

HOW IT WAS MADE

4.9 is the mean of 54 numbers

4.4 is the mean of 1,163. Both are means, computed the same way. They are not equally trustworthy, and nothing on the screen tells you that.

WHAT IT CANNOT SAY

4.9 cannot tell you the food is good

54 people could all be friends of the owner. Maybe it is not as trustworthy if the number of reviews is small — it has a name: sample size (cỡ mẫu).

READING A NUMBER

Puzzle 2, second answer — who writes a review at all?

Have you ever rated a restaurant that was simply 'fine' or 'OK'? Did you write the review?

HOW IT WAS MADE

Only customers who chose to write are counted

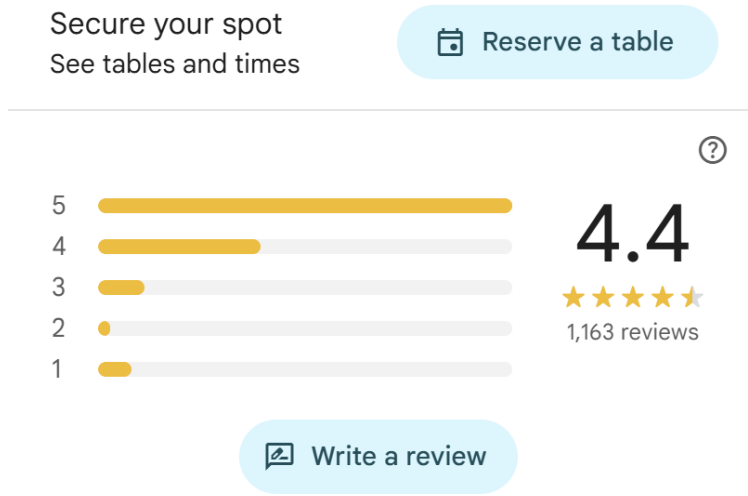
People write when they are **delighted**, or when they are **angry**. Nobody sits down to record that a meal was acceptable.

WHAT IT CANNOT SAY

The ratings are not the opinions that exist

This is why rating distributions on every platform are J-shaped, not bell-shaped.

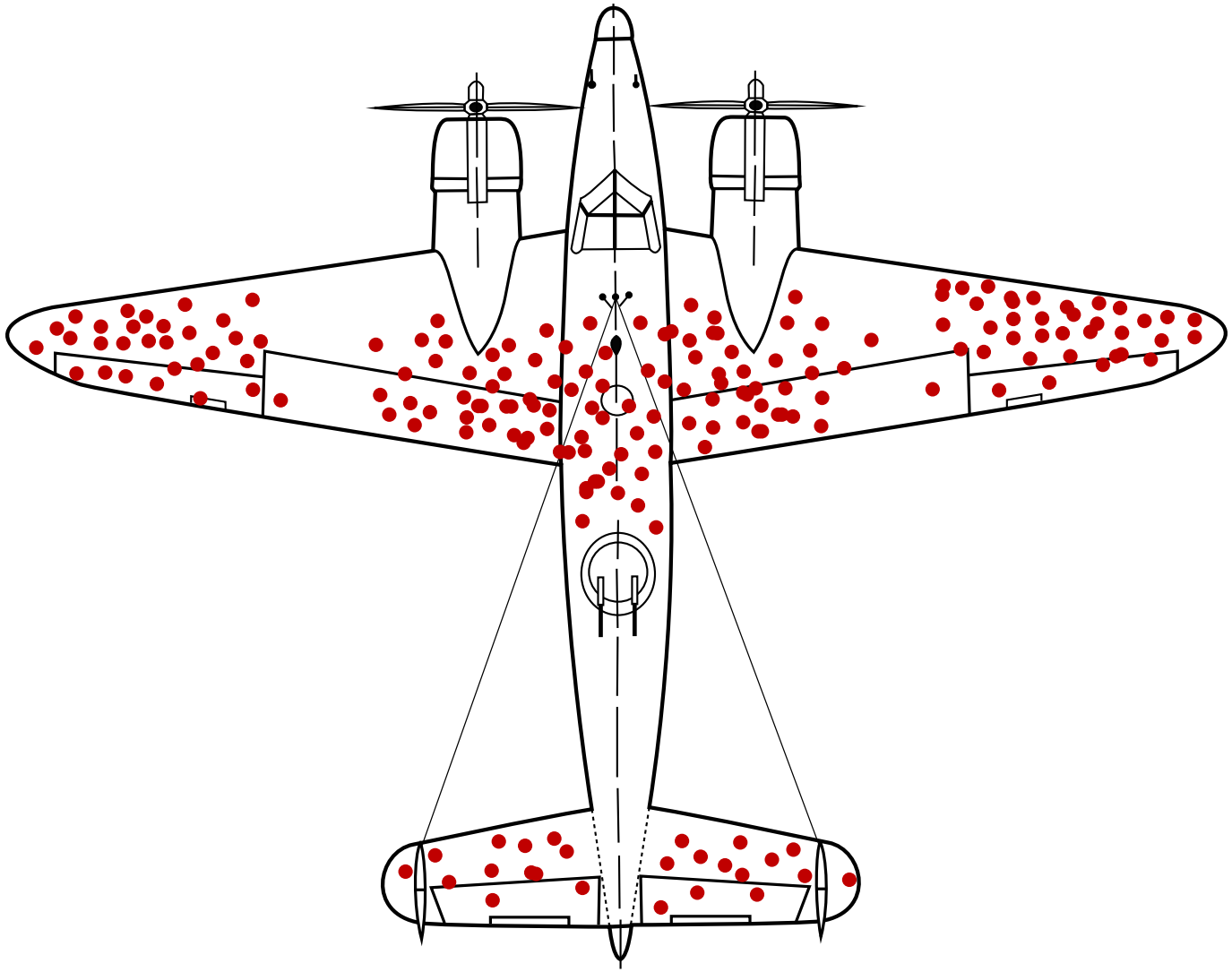




READING A NUMBER

Puzzle 3 — where do you add the extra armour?

WWII. Engineers record bullet holes on bombers that return to base. Holes cluster on wings and fuselage; cockpit and engines are almost clean. Your budget covers one area.



READING A NUMBER

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HOW IT WAS MADE

The data came from aircraft that came back

An aircraft hit in the engine did not come back, so nobody recorded its holes. The clean areas are clean precisely because those hits are fatal.

WHAT IT CANNOT SAY

Not where aircraft get hit — where survivors get hit

Abraham Wald's answer: armour the cockpit and the engines, the areas with no holes at all.

The question to ask of every dataset: who is missing from the data?

READING A NUMBER

Puzzle 4 — Should we trust him to take the penalty?

Elite footballers convert around 70-75% of penalties





READING A NUMBER

Now apply it to our own dataset

One dataset for all 14 sessions: Vietnamese listed firms, HOSE and HNX, one row per firm per year.

HOW IT WAS MADE

A firm is here because it is listed

Listing requires size, audited accounts, and a willingness to disclose. Nothing else gets in.

WHAT IT CANNOT SAY

Who is missing

Firms delisted after losses · bankrupt firms · small and household firms · firms that never wanted to disclose.

The data you can see is not the data that exists.

READING A NUMBER

Back to GDP vs Average income

Neither one is wrong. They just measure two different things.

	A — GDP per capita	B — Average income per capita
What is counted	Total value added of the whole economy	Money households actually receive
Includes	Corporate profit, taxes, depreciation, foreign investors' share	Wages, business income, transfers
How it is produced	National accounts, compiled from reports	A sample of households is interviewed
Whose money is it	Nobody's in particular	A household's
Verdict	Correct	Also correct — just a different question

READING A NUMBER

Three things follow from the examples

DEFINITION

Neither number means anything without its definition. A number without a definition is not information — it is only a string of digits

SAMPLING ERROR

A number produced by a survey carries sampling error. Phase 2 of this course, Sessions 4 to 6, exists entirely because of that

ASKING THE RIGHT QUESTION

The answers are derived from the question you are actually asking, not from the number

Your thesis says 'income of Vietnamese people'. Which number do you use?

READING A NUMBER

Numbers in a small group

DESCRIPTIVE

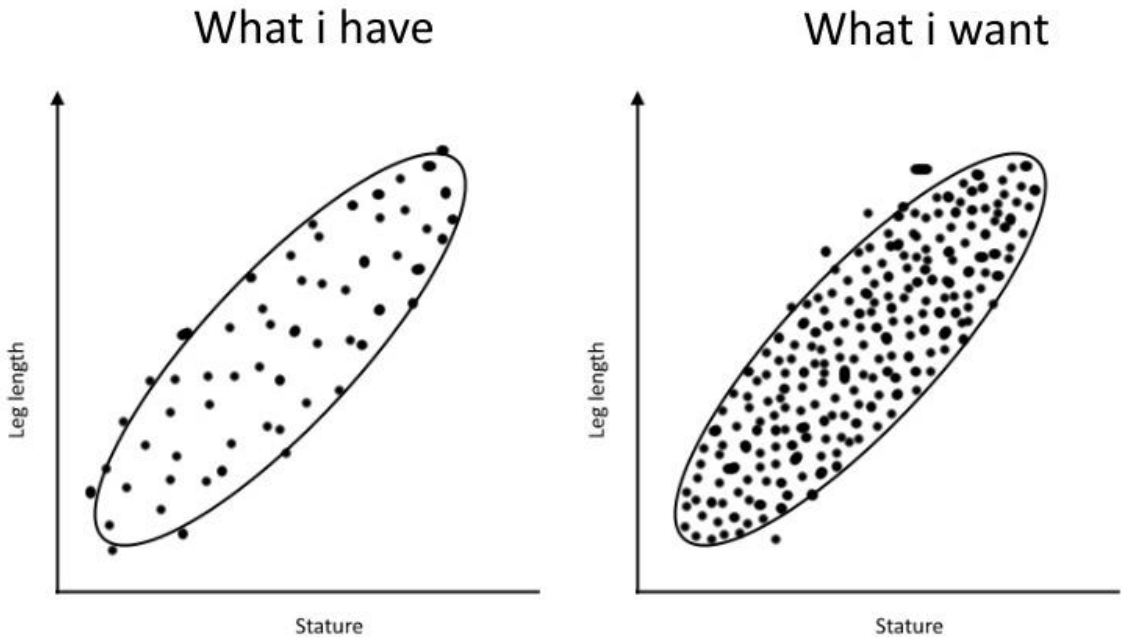
Describing what you have in your hands

The group is closed and you know every member of it. No uncertainty is involved at all.

INFERENTIAL

Talking about what you do not have

You hold a sample and want to say something about a bigger population. Uncertainty is unavoidable — and it can be measured.



READING A NUMBER

Descriptive or Inferential?

1

The average mark in this class is 7.2

DESCRIPTIVE — we hold every mark in the class.

2

VJCC students perform above average

INFERENTIAL — a claim about students we did not measure.

3

60% of firms in our dataset made a profit

DESCRIPTIVE — the dataset is the whole group being described.

4

60% of Vietnamese firms are profitable

INFERENTIAL — and, as we saw, not supportable here

Statements 3 and 4 differ by four words: 'in our dataset'.

READING A NUMBER

“Thống kê” in English meanings

Vietnamese uses 'thống kê' for both; English separates them by the article.

A STATISTIC

'The average mark in this class is 7.2' — one number computed from a group of data.

STATISTICS

'The Department of Statistics' — the name of the discipline.

'20% of this class received an A last semester.'

What can you infer from this information about the situation of the class in the last semester? This semester?

HOW 20% WAS CALCULATED: Count the A grades, divide by the number of students

8 out of 40. The counting is finished. Every one of those 40 students has a final mark and we know all of them.

There is no uncertainty here at all

This is pure description of a closed group. There is no probability anywhere in this sentence

So you have a 20% chance of an A this semester?

Yes or no? Vote before the reveal.

A rate for a group is not a probability for an individual

You are not a student drawn at random from last semester's class. Your mark depends on how you study, on this year's exam, and on how it is graded



THE MEANING OF PROBABILITY

A 70% penalty conversion rate

Someone counted the penalties he took and the ones he scored. 70% is the ratio of those two counts — and that is all it is.

1

Not the next kick

The next kick goes in or it does not. No kick goes in 70%. The number only exists across many repetitions.

2

Not from 10 kicks

7 out of 10 and 140 out of 200 are both '70%' and are not remotely the same evidence.

3

Not in every situation

Which goalkeepers? Which foot? Minute 90 of a final, or a training session? The rate averages over conditions that differ.

4

Not a debt owed to him

He has just missed two in a row — he is not 'due'. The ball does not remember anything.

How can we calculate the rate?

The sentence	What it actually is	Uncertainty?
20% of this class received an A	Description of a closed group	None
This player converts 70% of penalties	Long-run frequency, silent on the next event	Yes, next event
63% of Vietnamese support policy X	Inference from a sample to a population	Yes, measurable
40% chance of rain this afternoon	Output of a model — nothing was counted	Yes, hard to check

CALCULATION OF THE STATISTICS

Number 1 — '60% chance of rain this afternoon'

WHAT DOES IT MEAN?

Not 60% of the city, not 60% of the afternoon

It means: across all past days when the atmosphere looked like today, it rained on about 60% of them.

- It cannot be judged by today — if it stays dry the forecast was not 'wrong'
- The only test: take the 100 days they said 60%, and check whether it rained on about 60



CALCULATION OF THE STATISTICS

Number 2 — 'Your driver will arrive in 5 minutes'

Did the app count that, or guess it?

HOW IT WAS MADE

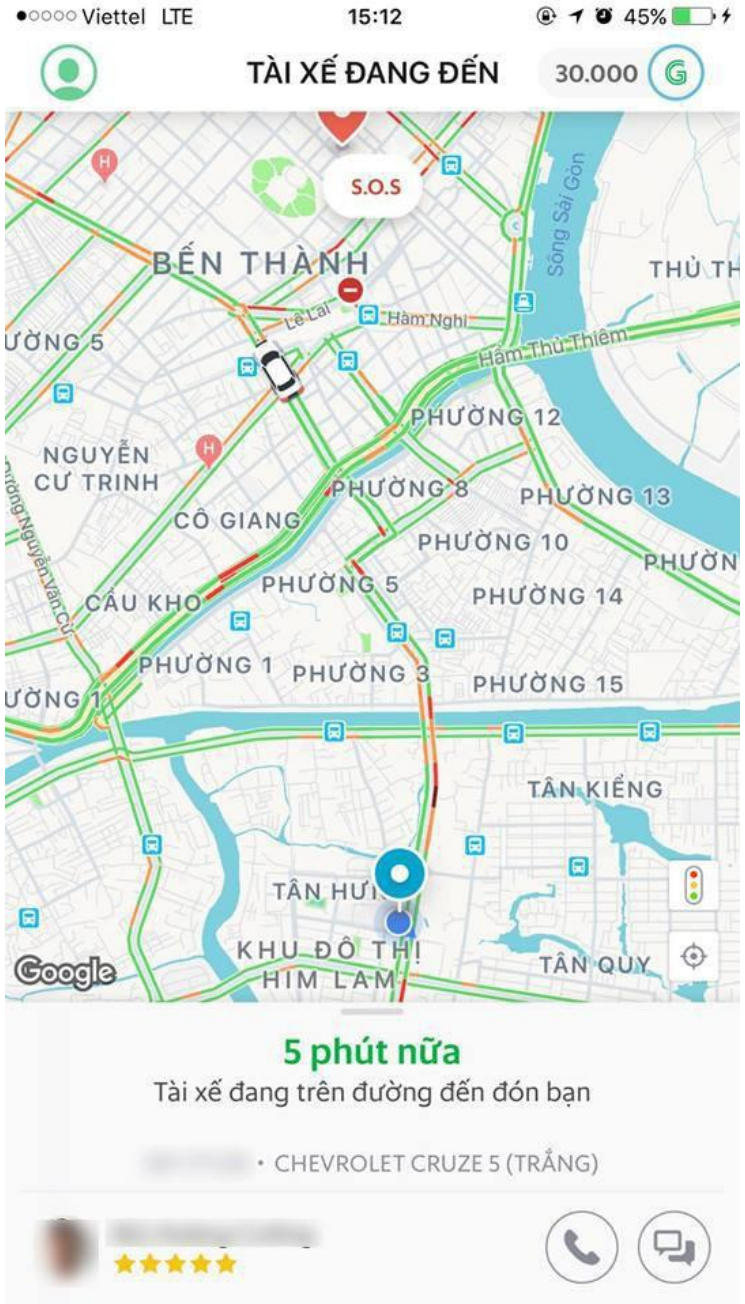
Guessed, properly — it is a model output

Fed by traffic, distance, driver behaviour and time of day. Nothing was counted. There is no '5' anywhere in the data.

WHAT IT CANNOT SAY

The honest answer may be 3 to 12 minutes

The app shows a single number because a single number is reassuring and a range is not.



CALCULATION OF THE STATISTICS

Number 3 — Vietnam's unemployment rate, 2025

Look at the situation what you see around you? Make a guess

2.22%

Working-age population

To be counted as unemployed you must be actively looking for work.

9.04%

Young people, aged 15–24

Same country, same year, four times the rate. The group you slice produces the number.



Vietnamnet global

Unemployment rate drops to 2.22% in Vietnam's first half of 2025

06/07/2025 12:26 (GMT+07:00) Follow VietNamnet on Google News

Vietnam sees rising employment and income growth, with strong labor market gains.



Vietnam+ Tiếng Việt English Français Español 中文 Русский

SOCIETY

Youth joblessness remains stubborn despite labour gains

The unemployment rate of young people aged from 15 to 24 was 9.04%, higher than previous quarter and same period previous years.

Someone who has stopped looking is not unemployed — they have left the labour force.

Source: Vietnamnet

Source: Vietnamnplus

CALCULATION OF THE STATISTICS

Number 4 — 'You are in the top 0.5% of this artist's listeners'

Top 0.5% compared with whom?

HOW IT WAS MADE

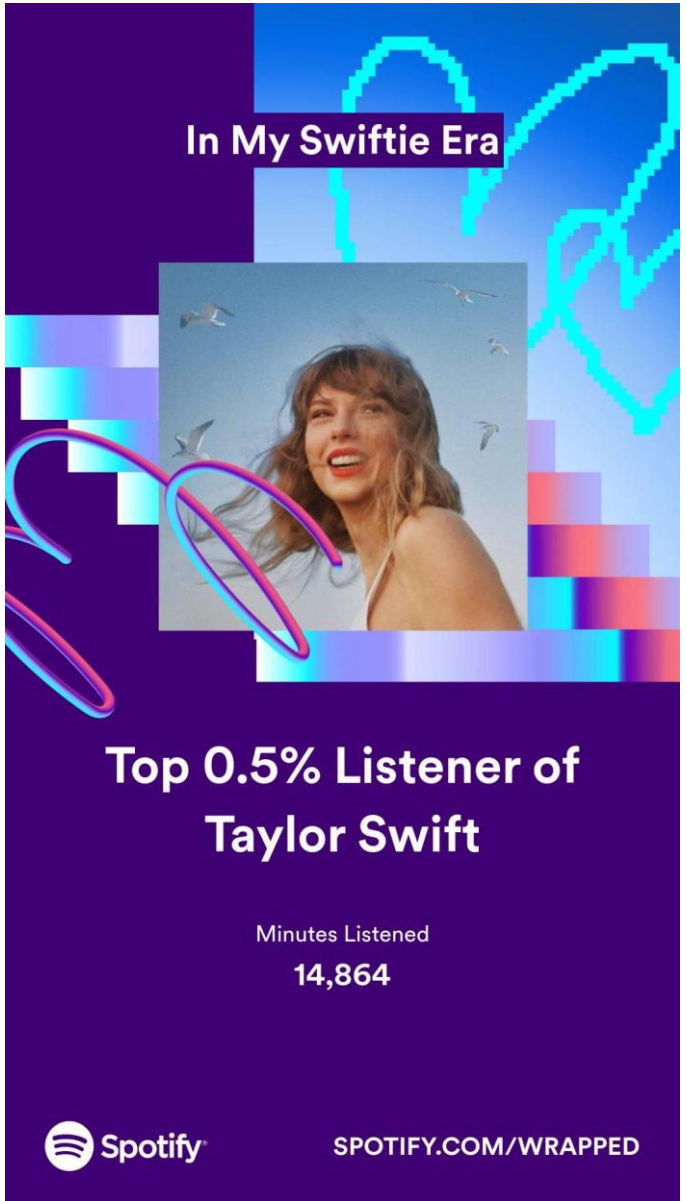
Everyone was ranked by listening time

You fell in the highest 0.5% of that ranking. This is a percentile (phân vị) — a position, not a quantity.

WHAT IT CANNOT SAY

It cannot tell you how much you listened

You could be top 0.5% on four hours or on four hundred. Change the comparison group — worldwide, or Vietnam only — and your number moves without you doing anything.



CALCULATION OF THE STATISTICS

Number 5 — who decided 'điểm chuẩn 27.9'?

Which committee met and voted on that number? Nobody did.

1.7	NTHKD07	CT ĐHNNQT Kinh doanh số toàn cầu	27.9
		CT ĐHNNQT Kinh doanh quốc tế theo mô hình tiên tiến Nhật Bản	

1

Applicants are ranked

By score, from the top down. No threshold exists yet.

2

Places are filled

From the top of that ranking until the places run out.

3

26.5 appears

It is the score of the last person admitted — an output, not a standard.

4

Next year it moves

More applicants, fewer places, or an easier exam all move it.

So '27.9 this year versus 26.7 last year' does not mean this year was harder.

CALCULATION OF THE STATISTICS

Number 6 — '95% of our graduates were employed within a year'

Who answered that survey?

HOW IT WAS MADE

Computed over the people who replied

A survey was sent to graduates. The 95% is a statistic about respondents, not about the graduating class.

WHAT IT CANNOT SAY

Someone still unemployed is less likely to reply

Same family of problem as the restaurant ratings on and the returning bomber. Non-response bias

CALCULATION OF THE STATISTICS

Number 7 — what counts as one video view?

PLATFORM A

A view is counted from the first frame. Scrolling past your video at speed still registers.

PLATFORM B

A view requires three seconds of watching. Same audience, visibly smaller number.

PLATFORM C

A view requires most of the video. '1 million' here and '1 million' above are not the same event, and cannot be compared.

The number is not lying. The definition is doing the work.

CALCULATION OF THE STATISTICS

Three questions. Every number. Forever.

1

What does it count, under whose definition?

2

Is it describing a closed group, or predicting something that has not happened?

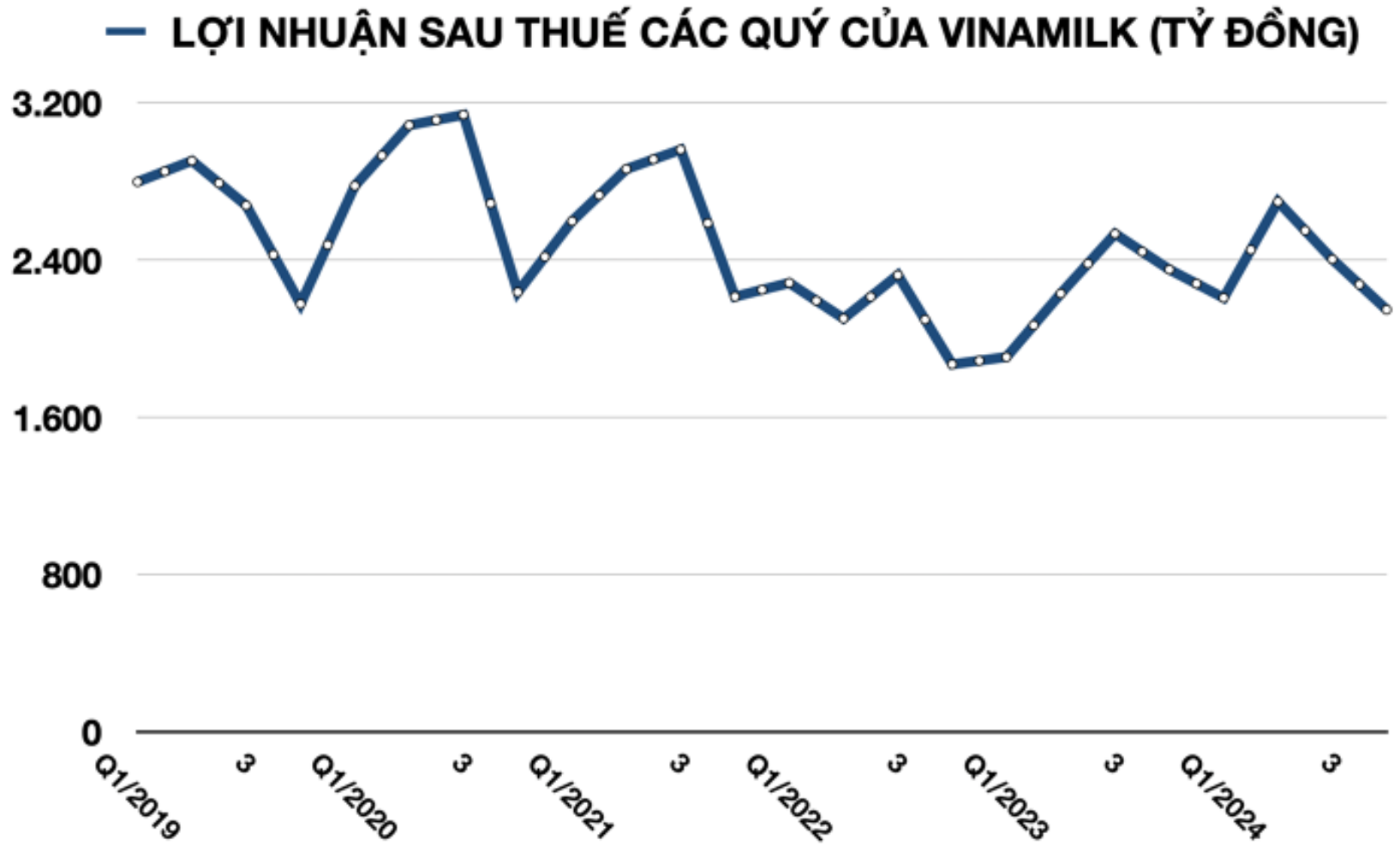
3

Who is missing from the data that produced it?

FROM DATA TO DECISION

What is this chart showing?

Can you guess what is the unit? Billion VND? Million USD? Number of employees?



FROM DATA TO DECISION

Comparison, then judgement

COMPARISON

'Flat for five years while the sector grew'

We compared the number with itself over time and with peers. Nothing new was measured — only compared.

JUDGEMENT

'So delay the new plant'

A human weighed the evidence against goals, risk appetite, and things not in the data at all. The number did not produce this sentence.

Data supports the decision. It does not make it, and it does not carry the blame.

FROM DATA TO DECISION

You just climbed a ladder

DIKW — we derived it first and named it second. That order is deliberate.

DATA

VNM · Q1-2024 · a number

To climb, add: labels, units, entity, period.

INFORMATION

'Net revenue was 2,400 billion VND'

To climb, add: comparison over time and against peers.

KNOWLEDGE

'Flat while the sector grew'

To climb, add: judgement, goals, risk, accountability.

WISDOM

'So delay the new plant'

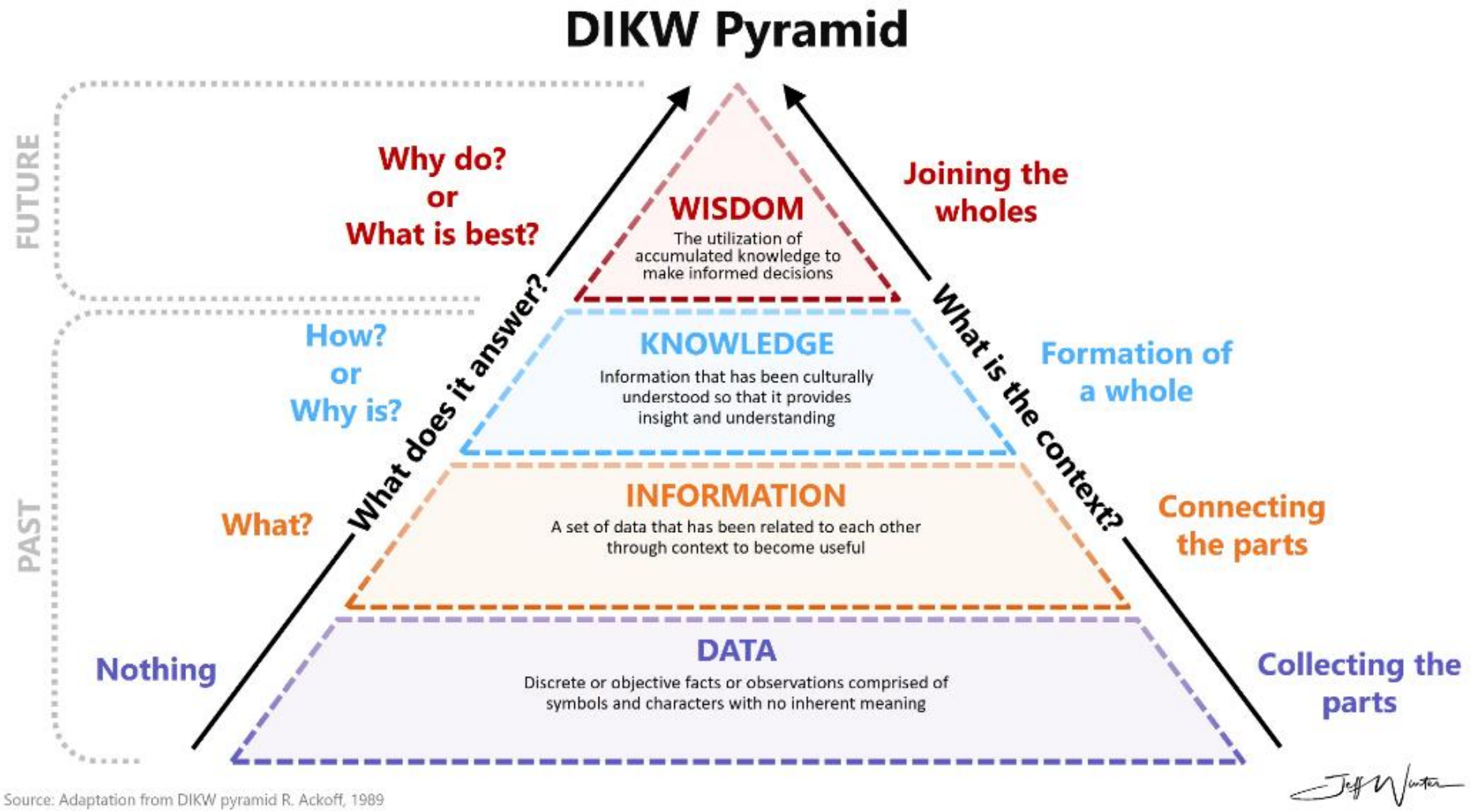
A person owns this rung, and answers for it.

Is DIKW a law? And where does an AI model sit on this ladder? No agreed answer for either.

FROM DATA TO DECISION

You just climbed a ladder

DIKW — we derived it first and named it second. That order is deliberate.



Source: Adaptation from DIKW pyramid R. Ackoff, 1989

WHERE DATA COMES FROM

Primary and Secondary data

PRIMARY · SƠ CẤP

You designed the measurement yourself

You know the definitions because you wrote them. Expensive, slow, and yours.

SECONDARY · THỨ CẤP

Someone else measured, for their own purpose

Under their own definitions. Cheap, fast, and you must reconstruct what they did before you may use it.

Almost all of your data will be secondary. So the skill is not measuring — it is reading how someone else measured.

WHERE DATA COMES FROM

Activity 1 — trace the source

Your group is given one economic indicator. Find it, and report where it really came from.

THỜI LƯỢNG

18 phút — 8 phút tra cứu theo nhóm, 10 phút báo cáo trước lớp.

SỐ LƯỢNG

Báo cáo đủ 5 mục: giá trị · nguồn chính xác · ai sản xuất · sơ cấp hay thứ cấp với nhóm bạn · tin đến mức nào và vì sao.

RÀNG BUỘC

Phải trích được định nghĩa của chỉ số, bằng đúng câu chữ của cơ quan sản xuất. Một con số không có định nghĩa được không điểm.

Một dòng viết lên bảng cho nhóm mình, giữ nguyên đến hết buổi.

WHERE DATA COMES FROM

Activity 1 — group assignments

15 minutes · groups of 4–5 ·

Report all 5: (1) value · (2) URL/source · (3) producing organisation · (4) primary/secondary for you · (5) trust level.

Group	Find this	Likely producer
1	CPI for the most recent month	National statistics office
2	The refinancing rate currently in force	State Bank of Vietnam
3	VN-Index at the last close, and the number of listed tickers on HOSE	HOSE
4	GDP per capita: Vietnam, Thailand, Indonesia	World Bank, WDI

WHERE DATA COMES FROM

What all four numbers have in common

Finding the definition is finding the data.

Every one of these numbers has an organisation accountable for it and a written definition you can read. And for your group, all four are secondary — none of you measured anything, so you can only report who did, and how.

WHERE DATA COMES FROM

The producer itself is metadata

HOW IT WAS MADE

Statistical agencies are government bodies

Their names and mandates change when the government reorganises. Vietnam's statistics agency was restructured in 2025 — the name you learned at school may no longer be correct.

WHAT IT CANNOT SAY

A wrong agency name cannot be verified

Check the producer's current legal name at the moment you cite it, not from memory. Current name, decree number and date are on today's reference sheet.

WHERE DATA COMES FROM

Measurement error is a property of the method

A survey asks: 'what was your income last month?'

STREET VENDOR

Estimates. Income varies day to day and nothing was written down, so the answer is a recollection, not a record.

HIGH EARNER

May under-report, for reasons that have nothing to do with the survey. The error here runs in one predictable direction.

SALARIED EMPLOYEE

Answers precisely. Three respondents, three error sizes — and the resulting average cannot be more accurate than the question that produced it.

The fix is not to distrust the data. It is to know which direction the error runs, and say so.

Excel or Python?

Task	Excel	Python
How many firms and how many years are in this file?	15 seconds. Everyone can do it, and you see every cell.	3 lines. Also fast.
Now do the same for 40 files, one per sector.	Open 40 times, paste 40 times. One mistake and nobody notices.	Change one line.
Session 13: VIF, robust standard errors, logistic regression.	Not available.	Available.
So	Excel to SEE the data.	Python to REPEAT it, and to go further.

Programming Time!

YOU WILL BE ASKED TO

Run a prepared notebook and read the output

Say in plain economic language what it means and what it does not mean. Build tables and figures that meet publication standards.

YOU WILL NOT BE ASKED TO

Write code from a blank file

No memorising formulas. No deriving proofs. Every notebook is provided and already runs.

The only requirement is that you understand what you are running.

OPENING THE FILE

Before we open the file — how many companies?

Write your answer down first. Then we count, live.

[N_ROWS]

rows in the file

df.shape — this is the number almost everyone answers with.

[N_FIRMS]

unique tickers

df['ma_ck'].nunique() — the actual number of companies.

[N_YEARS]

unique years

df['nam'].nunique() — and firms × years is roughly the row count.

OPENING THE FILE

So what is one row?

HOW IT WAS MADE

One row = one firm, in one year

The unit of observation is firm × year. This structure is called panel data. The same ticker appears once for every year it was listed.

WHAT IT CANNOT SAY

Row count cannot tell you firm count

You have to look at vertically. And 'the average of this column' is now an ambiguous instruction.

OPENING THE FILE

You select the whole ROA column and press AVERAGE

What did you just compute — the average ROA of a firm, or of a firm-year? Are those the same number?

HOW IT WAS MADE

Every row got one vote

A firm listed for 10 years is counted 10 times. A firm listed for 2 years is counted twice. Your 'average firm' is silently weighted toward firms listed longest.

WHAT IT CANNOT SAY

It is not the average firm

Excel will still return a number. It will still have four decimal places. It will still be the wrong answer to the question you asked.

Raw data is not information until you know what one row is.

OPENING THE FILE

Activity 2 — can our dataset answer this?

Five statements. For each one decide YES or NO — can this dataset support it? — and give the reason.

Statement	Verdict	Why
The average ROA of Vietnamese firms is 7%		
Listed real-estate firms carry more debt than listed consumer-goods firms		
Borrowing more reduces ROA		
In 2020, listed firms had lower ROA than in 2019		
80% of Vietnamese firms are profitable		

SUMMARY

DEFINITION

What does this number count, and under whose definition? Phases 1 and 4 keep returning to this.

CLOSED OR PREDICTED

Is it describing a group I hold, or predicting one I do not? That split is Phase 2 and Phase 3.

WHO IS MISSING

Who is absent from the data that produced it? This one never stops being the right question.